TCS CORPORATE WEBSITE

Modal Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-08
Documents	1
Classification	INTERNAL — classification to be confirmed

Generated from the Markdown sources of docs/handover (branch migrated-page-content-edam); do not edit this PDF.

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Modal Block — Developer Guide

Attribute	Value
Document ID	DEV-B-modal
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the Modal block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/modal/ and its components modal
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	2026-11-08 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/modal/README.md, blocks/modal/_modal.json, blocks/modal/modal.css, blocks/modal/modal.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	/content/tcs-corp/design-block-master/en/block-library/modal, /content/tcs-corp/design-block-master/en/block-library/modal/auto-linked-modal, /content/tcs-corp/design-block-master/en/block-library/modal/fragments

1. Summary

Attribute	Value
Block	Modal
Folder	`blocks/modal`
Type	Independent
Category	interactive
Status	Active
Purpose	Opens a page's content in a dialog (button or auto-linked /modal/ links)
Variations (summary)	Trigger button; auto-linked
Children	—
Code size	276 lines of JS and CSS

2. Overview

Business purpose. Show supplementary content (details, terms, a short form, a video) without leaving the page; visitors return to the same place when they close it.

User experience. A primary button (for example "View our awards"); selecting it dims the page and opens a white dialog, centred, at most 56.25rem wide, with a close button at the top right; long content scrolls

inside the dialog while the page behind stays still. The close button, Escape or a click on the backdrop closes it and focus returns to the button.

Templates / page types. Any page.

Data source. The fragment page named in Modal Content, fetched (`{path}.plain.html`) and decorated each time the dialog opens.

Related blocks. Fragment (same loader, inline instead of a dialog), the Video block (`blocks/video`, Play in a Pop-up uses `createModal`), Form Variations (a form on a modal page), Button Block (a button linking to a `/modal/` page opens a modal too).

When to use. Short, secondary, self-contained content.

When not to use. Primary or indexable content (it is loaded from another page on demand), long articles, multi-step processes, content that must be shared by URL; automatic pop-ups on page load aren't implemented.

Realistic use cases. "View our awards" (block library `block-library/modal/modal-block`, fragment `modal/fragments/recognition`); "Read the programme terms" with long content (`modal-long-content`); "Contact our team" with a form (`modal-with-a-form`); a privacy notice linked from running text.

3. Source files

File	Link
`README.md`	GitHub
`_modal.json`	GitHub
`modal.css`	GitHub
`modal.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
`modal`	Modal	`core/franklin/components/block/v1/block`	Modal	`modal`	—

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 Modal (`modal`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Modal Content	`link`	aem-content	No	—	—	—
Content	Button Text	`linkText`	text	No	—	—	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import
../fragment/fragment.js
../../scripts/aem.js
../../scripts/scripts.js
../../decorators/button-decorator.js
../../decorators/icon-button-decorator.js
../../decorators/data-layer-decorator.js
../../utils/analytics.js
../../utils/environment.js
../../utils/ids.js
../../utils/placeholders.js

7. For Developers

```
blocks/modal/
├── README.md
├── _modal.json
├── modal.css
└── modal.js
```

Decorator/Utility	Purpose	Execution or dependency point
blocks/fragment/fragment.js	—	modal.js: { loadFragment }
scripts/aem.js	Edge Delivery core library (block loading, CSS, metadata, optimized pictures)	modal.js: { buildBlock, decorateBlock, loadBlock, loadCSS, toClassName, }
scripts/scripts.js	Page decoration (moveInstrumentation and shared page helpers)	modal.js: { moveInstrumentation }
decorators/button-decorator.js	decorateButton (option newTabHint: a link opening a new tab says so), createButton, setButtonDisabled, setButtonAlignment, removeButtonDecoration	modal.js: { decorateButton }
decorators/icon-button-decorator.js	createIconButton	modal.js: { createIconButton }
decorators/data-layer-decorator.js	trackPage, trackBlocks, trackInteractions, componentIdOf (Adobe Client Data Layer for every block, Core Components schema, read by Adobe Tags / Web SDK; see "Data layer" below)	modal.js: { componentIdOf }
utils/analytics.js	pushToDataLayer, pushComponents, pushComponentEvent (Adobe Client Data Layer, Core Components schema; never throws)	modal.js: { pushComponentEvent }
utils/environment.js	prefersReducedMotion, scrollBehavior, isEditing, isInEditor, isAuthor (the page is rendered by AEM author, which has no query index)	modal.js: { isEditing }
utils/ids.js	nextInstance, uniqueId, createIdFactory (unique ids per block instance), availableId (an authored id kept, suffixed -2, -3, ... only when taken)	modal.js: { uniqueId }
utils/placeholders.js	getPlaceholders, loadPlaceholders, fillPlaceholder (UI text from the locale's placeholders, with English fallbacks)	modal.js: { getPlaceholders }

Input DOM. The Modal Content link with its collapsed Button Text (one `a[href]`, possibly in a `p.button-container`); the Additional tab as block classes.

`decorate(block)`. No link: empty block (edit mode: `p.modal-empty`). Else a `button` (text, `title`, `aria-haspopup`), `moveInstrumentation(link, button)`, `decorateButton(button, { view: 'button', type: 'primary' })`, click → `openModal(href, { opener: button, id: ${componentIdOf(block)}-dialog })`.

`openModal(fragmentUrl, { opener, id })` (exported). Path from an absolute URL; a `Set` of paths being opened ignores repeated clicks; `loadFragment(path)`; `null` → nothing; `createModal(fragment.childNodes, { id: id || modal-${toClassName(path)} })`, then `showModal(opener)`.

`createModal(contentNodes, { id })` (exported). Placeholders; `loadCSS('blocks/modal/modal.css')`; `dialog > button.close-button + div.modal-content`; the name; a `modal` block built with `buildBlock`, given `data-cmp-id`, appended to `main`, decorated and loaded, then filled with the dialog. Returns `{ block, showModal(from) }`. `showModal` stores the opener, starts `trackVisibleArea` (only in a frame: an `IntersectionObserver` on the document and a non-passive `wheel` listener, both removed on close), calls `dialog.showModal()`, pushes `cmp:show`, resets the scroll and adds `body.modal-open`. On `close`: stops tracking, removes `modal-open`, removes the block, focuses the opener.

Rendered DOM contract.

```
<div class="modal block" data-block-name="modal" data-cmp-id="anchor-modal-awards-dialog">
  <dialog aria-labelledby="modal-title-1" open>
    <button type="button" class="close-button button tertiary" aria-label="Close" title="Close">
</button>
    <div class="modal-content">
      <div class="section">…<h2 id="modal-title-1">Awards and recognition</h2>…</div>
    </div>
  </dialog>
</div>
```

Auto-link. `scripts/scripts.js` `autolinkModals(doc)` (lazy phase) imports `blocks/modal/modal.js` on the first matching click.

Error handling. A failed fetch resolves to nothing (no dialog, no message); a network error rejects inside `openModal` (the `finally` clears the guard).

8. Extending the Block

- **A field (e.g. a button style).** Add it to `_modal.json` (Additional tab last), `npm run build:json`, `npm run lint`. The block isn't key-value: each dialog tab adds an empty row, so read by `data-aue-prop` in the editor and skip empty rows; a field ending in Text, Title, Type or Alt collapses into its base field (use e.g. `buttonVariant`, as the Button Block does).
- **Rendering.** Keep the opener a `<button>` with `aria-haspopup`; pass a stable `id` to `openModal` / `createModal` from new callers.
- **Responsive.** `modal.css` uses the 900px breakpoint and `--modal-top` / `--modal-height`.
- **Accessibility.** Keep `showModal()`, the name, focus return and the close button.
- **Data layer.** Keep `cmp:show` here and `cmp:hide` from the central `close` listener.
- **Tests.** No unit test covers the block; `test/motion/motion.test.mjs` checks it has no animation fields.
- **Docs and modules.** Update this README (`npm run build:docs`) and the block library modal pages; `npm run build:modules` after changing imports.

9. Responsive Behavior

Viewport	Layout	Image behavior	Interaction behavior
Below 900px	Dialog `calc(100vw - 3rem)` wide, gap 1.5rem above and below, content padding 0 1.5rem 1.5rem	As the content's blocks	Opener button; scroll inside the dialog; Escape, close button, backdrop close
900px and above	Dialog `calc(100vw - 4rem)` wide, at most 56.25rem, gap 2.5rem, content padding 0 2rem 2rem	As above	As above

The dialog is centred in the visible area (`--modal-top`, `--modal-height` in a frame, else the viewport) and never taller than it.

10. Accessibility

- **Semantics.** Native `<dialog>` opened with `showModal()` (modal, the rest of the page inert); named by `aria-labelledby` (first heading) or `aria-label` "Dialog".
- **Opener.** A `<button>` with `aria-haspopup="dialog"` (a button, not a link, so it works in Universal Editor preview too).
- **Focus.** On open the browser focuses the first focusable element (the close button); on close focus returns to the opener passed explicitly (works in Safari, which doesn't focus clicked buttons).
- **Close.** Close button (icon button, `aria-label` and `title` "Close"), Escape (native), a click on the backdrop outside the dialog's box.
- **Scroll.** The page is locked while open; the dialog scrolls; in a frame (Universal Editor) the dialog is kept in the visible area and hidden until measured.
- **Reduced motion.** No animation in the block.

Author responsibilities: a heading at the top of every modal page, a descriptive Button Text, one purpose per dialog.

Developer responsibilities: keep `showModal()`, the naming, the explicit opener and the placeholders; never open a dialog without a visitor action.

Key	Expected behavior
Enter / Space	On the opener: opens the dialog; on the close button: closes it
Tab / Shift+Tab	Moves within the dialog (the page behind is inert)
Escape	Closes the dialog; focus returns to the opener

11. Data Layer and Analytics

- **Opener component.** The authored block (`modal-{n}`, or its Anchor ID), `@type` `tcs-corp/components/modal`, added after the main sections load.
- **Dialog id.** Stable: `{opener id}-dialog` for the Modal block, `modal-{path}` (`toClassName`) for auto-linked links; Video pop-ups get `modal-{n}`. The dialog block is created at opening and removed at closing; it is added to the component state only if it is open during the delayed `trackBlocks` pass.
- **Events.** `modal.js` pushes `cmp:show` on open (`pushComponentEvent`); `decorators/data-layer-decorator.js` pushes `cmp:hide` on the dialog's `close` event (any way of closing). Clicks are captured centrally.

Trigger	Data-layer event	Required properties	Analytics behavior
Click on the opener	`cmp:click`	`eventInfo.path` `component.{opener id}`, `interaction` "button", `dc:title` (button text), `breakpoint`	Opener use
Dialog opened	`cmp:show`	`eventInfo.path` `component.{dialog id}`, `interaction` "dialog"	Dialog impression

Trigger	Data-layer event	Required properties	Analytics behavior
Dialog closed (button, Escape, backdrop)	`cmp:hide`	`eventInfo.path` `component.{dialog id}`, `interaction` "dialog"	Dialog close
Click on the close button	`cmp:click`	`interaction` "button", `dc:title` "Close"	Close intent
Click on a link in the dialog	`cmp:click`	`xdm:linkURL`, `xdm:linkType`	Link from the dialog

The intended flow is Adobe Client Data Layer → Adobe Data Collection (Tags) → AEP Web SDK → Edge Network → Analytics / CJA; Tags and the Web SDK aren't configured yet (blocked: measurement plan). No consent mechanism is implemented. Build rules on `interaction: "dialog"`.

12. Performance Considerations

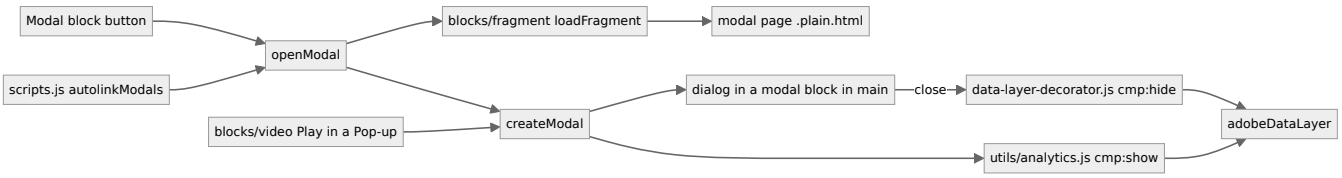
- The authored block only builds a button (eager or lazy with its section); the modal stylesheet, the placeholders and the page are loaded when the dialog opens.
- Auto-link: one document click listener from the lazy phase; `modal.js` imported on the first matching click.
- The page is fetched and decorated again at each opening (no caching in code); the dialog block is removed on close.
- In a frame only: an `IntersectionObserver` (101 thresholds) and a non-passive `wheel` listener while open, both removed on close.

13. Security and Privacy

- Only root-relative same-site paths are loaded (`loadFragment`); absolute URLs are reduced to their path.
- Auto-link checks only the path (`/modal/`), not the link's origin: an external link whose path contains `/modal/` would open the same path of this site instead of navigating.
- Content is the site's own HTML; no query strings, no personal data; form data in a dialog is handled by Form Variations.

14. For Architects

- **Role.** On-demand presentation of a content page in a dialog; a shared dialog service for blocks.
- **Content model.** A link to a page; the page is the content.
- **Rendering.** Client-side; content fetched and decorated at each opening; the dialog lives in a temporary block in `main`.
- **Dependencies.** `blocks/fragment`, `scripts/aem.js` (`buildBlock`, `decorateBlock`, `loadBlock`, `loadCSS`, `toClassName`), `scripts/scripts.js` (`moveInstrumentation`, `autolinkModals`), `decorators/button-decorator.js`, `decorators/icon-button-decorator.js`, `decorators/data-layer-decorator.js`, `utils/analytics.js`, `utils/environment.js`, `utils/ids.js`, `utils/placeholders.js`.
- **Data layer / Web SDK / Target.** `cmp:show` / `cmp:hide`; no Target.
- **Consent and identity.** None.
- **Caching.** The modal page through the CDN and browser cache; no caching in code.
- **Performance.** No cost until opened.
- **Security.** Same-site root-relative paths only.



15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
The button does nothing	Modal page not published, wrong path, or not a site path	Network: `{path}.plain.html` status	Publish / correct Modal Content	Dialog opens
No button	Modal Content empty, or Hide block on	Properties panel	Choose the page; switch Hide block off	Button visible
Button says "Open"	Button Text empty or equal to the URL	Properties panel	Enter a descriptive text	Correct label
A link opens a dialog (or navigates) unexpectedly	Its path contains `/modal/` (auto-link), or clicked before the lazy phase / in the editor	Check the URL	Move pages out of `modal` folders; use a Modal block in the editor	Intended behaviour on `.aem.page`
Screen readers say "Dialog" only	Modal page without a heading	Inspect `aria-label`	Add a heading	`aria-labelledby` set
Content unstyled or blocks missing	A block of the modal page failed	Console `failed to load …`	Fix the content; escalate block errors	Renders as on its page
No `cmp:show` / `cmp:hide`	Dialog didn't open; Tags not configured	`adobeDataLayer.filter((e) => e.eventInfo?.interaction === 'dialog')`	Fix the opening; configure Tags later	Events present

Evidence to collect: host page URL, Modal Content value, network status of the modal page request, console errors, the data-layer entries. No cookies or tokens.

16. Known limitations of the block

Limitation	Impact	Workaround	Planned correction
The Video pop-up player (`.modal .video-modal-player`, up to 75rem) is wider than the dialog (56.25rem) (F-51)	Player runs past the dialog edge at 1440px	—	Not recorded (F-51 open, P2)
The modal page is fetched at every opening	Repeated requests	—	Not planned
The opener is always a primary button (no style option, audit)	No secondary / tertiary opener	Use a Button Block linking to a `/modal/` page	Not recorded
Auto-link ignores the link's origin	External `/modal/` links don't navigate	Avoid such links	Not recorded
Auto-link starts in the lazy phase and not in Universal Editor (the editor navigates)	Early clicks navigate	Use a Modal block for editor previews	Not planned
A failed fetch shows no message	Silent failure	Publish the modal page	Not recorded

17. Change history of the block

Version	Change	Source	Date
Not recorded	README generated from verified implementation and approved handover documentation	Repository review	Not recorded
Handover 2.27	Block ecosystem audit: one dialog per double click (D-C-9), stable data-layer id from the opener / path (D-C-10), opener passed explicitly for focus return (D-C-11), editor hint without Modal Content (D-C-21)	docs/handover revision history; docs/audit/BLOCK-ECOSYSTEM-AUDIT.md	19 October 2026

Version	Change	Source	Date
Handover 2.29	Re-audit: the Video pop-up in the dialog corrected	docs/handover revision history	20 October 2026

18. Reference pages

Kind	Page	Live URL
Block library	/design-block-master/en/block-library/modal	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal
Block library	/design-block-master/en/block-library/modal/auto-linked-modal	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/auto-linked-modal
Block library	/design-block-master/en/block-library/modal/fragments	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/fragments
Block library	/design-block-master/en/block-library/modal/fragments/contact-form	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/fragments/contact-form
Block library	/design-block-master/en/block-library/modal/fragments/privacy-notice	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/fragments/privacy-notice
Block library	/design-block-master/en/block-library/modal/fragments/recognition	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/fragments/recognition
Block library	/design-block-master/en/block-library/modal/fragments/terms	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/fragments/terms
Block library	/design-block-master/en/block-library/modal/modal-block	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/modal-block
Block library	/design-block-master/en/block-library/modal/modal-long-content	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/modal-long-content
Block library	/design-block-master/en/block-library/modal/modal-with-a-form	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/modal/modal-with-a-form

19. Coverage of the required topics

Topic	Where
Purpose	Overview
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	MODAL-VARIATIONS.md
Responsive behaviour	Responsive Behavior
Accessibility and keyboard	Accessibility
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations

Topic	Where
Troubleshooting	Troubleshooting
Tests	MODAL-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

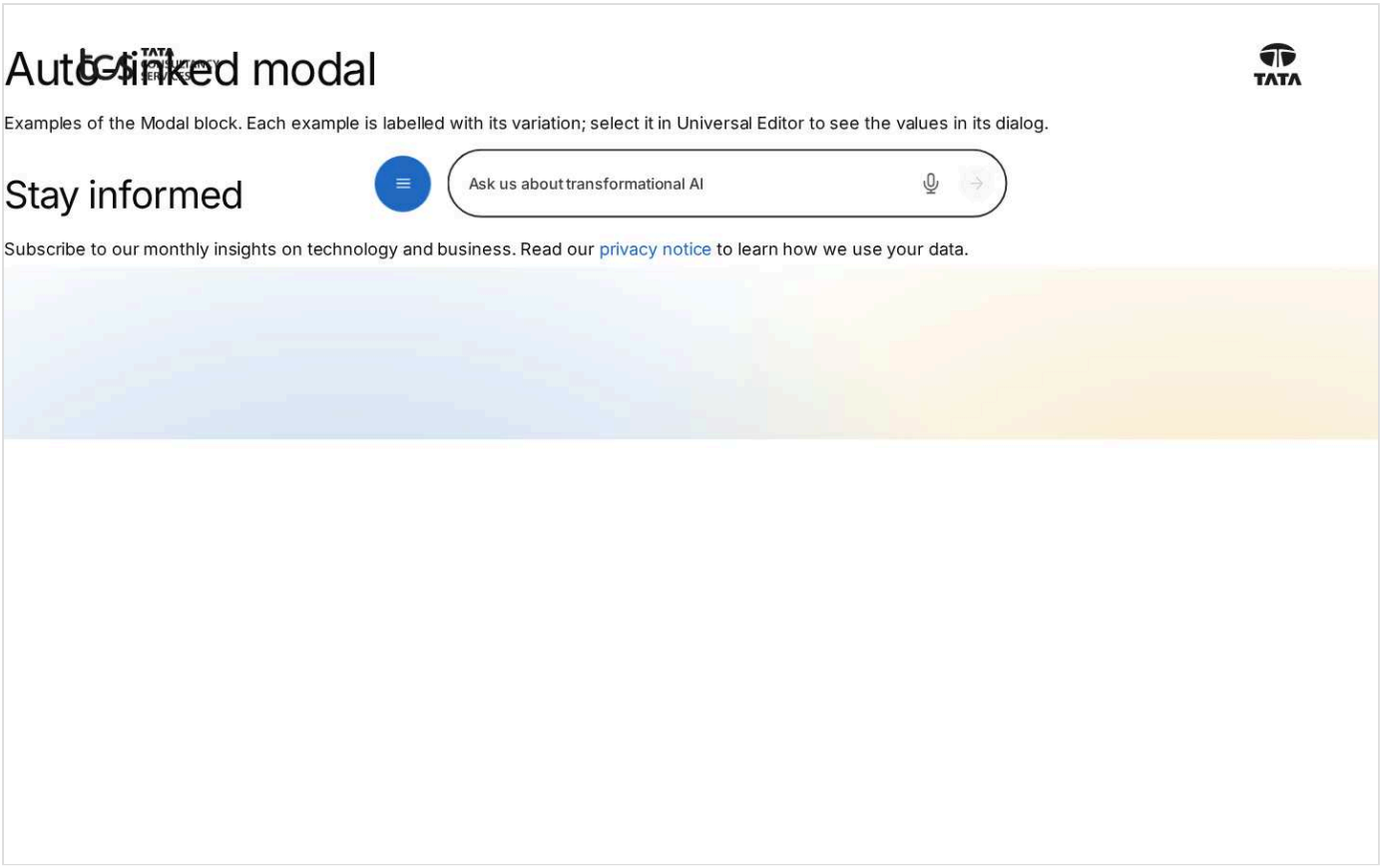


Figure: Modal block library page — desktop 1440 × 900 (/design-block-master/en/block-library/modal/auto-linked-modal, captured 2026-11-08).

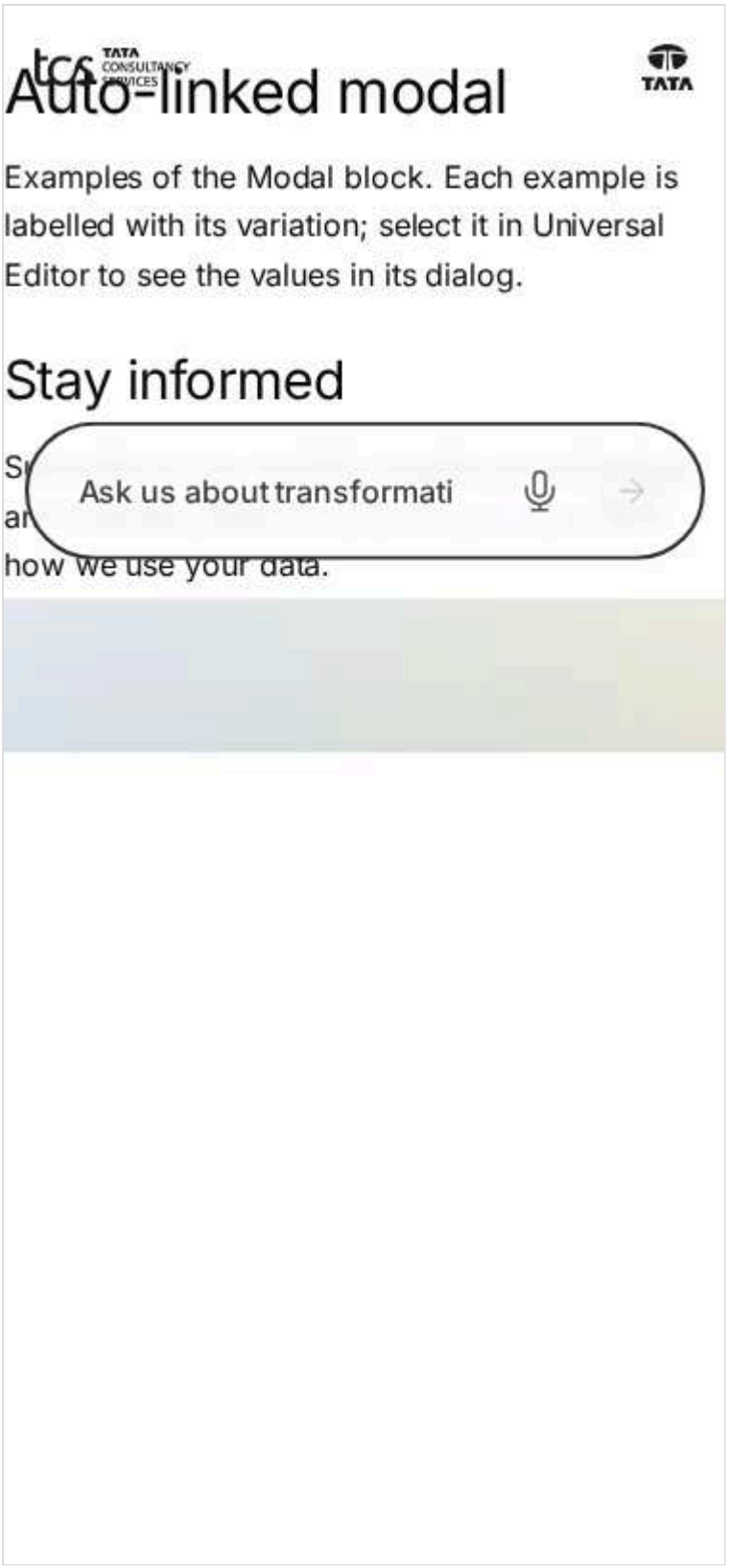


Figure: Modal block library page — mobile 375 × 812 (/design-block-master/en/block-library/modal/auto-linked-modal, captured 2026-11-08).

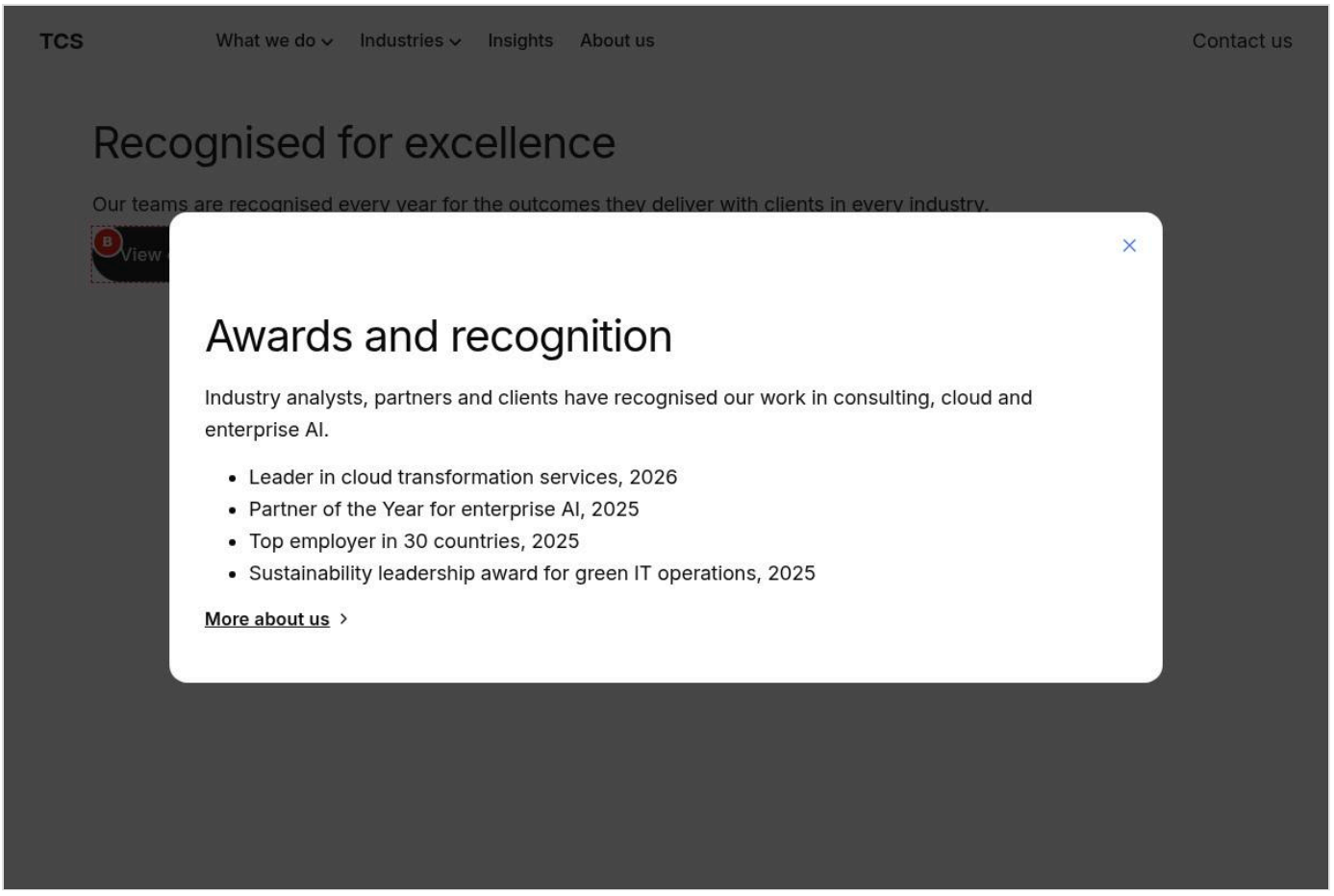


Figure: Desktop rendering — At 1200 px, after selecting View our awards: the page dimmed by the near-black 75 % backdrop and, centred over it, a white dialog 56.25rem wide with 1rem rounded corners, the close (X) button at the top right, the H2 Awards and recognition, the introduction, four bulleted awards and the link More about us, with 2rem padding. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/modal-desktop-view.jpg) (handover screenshot; rendering at the time it was taken).

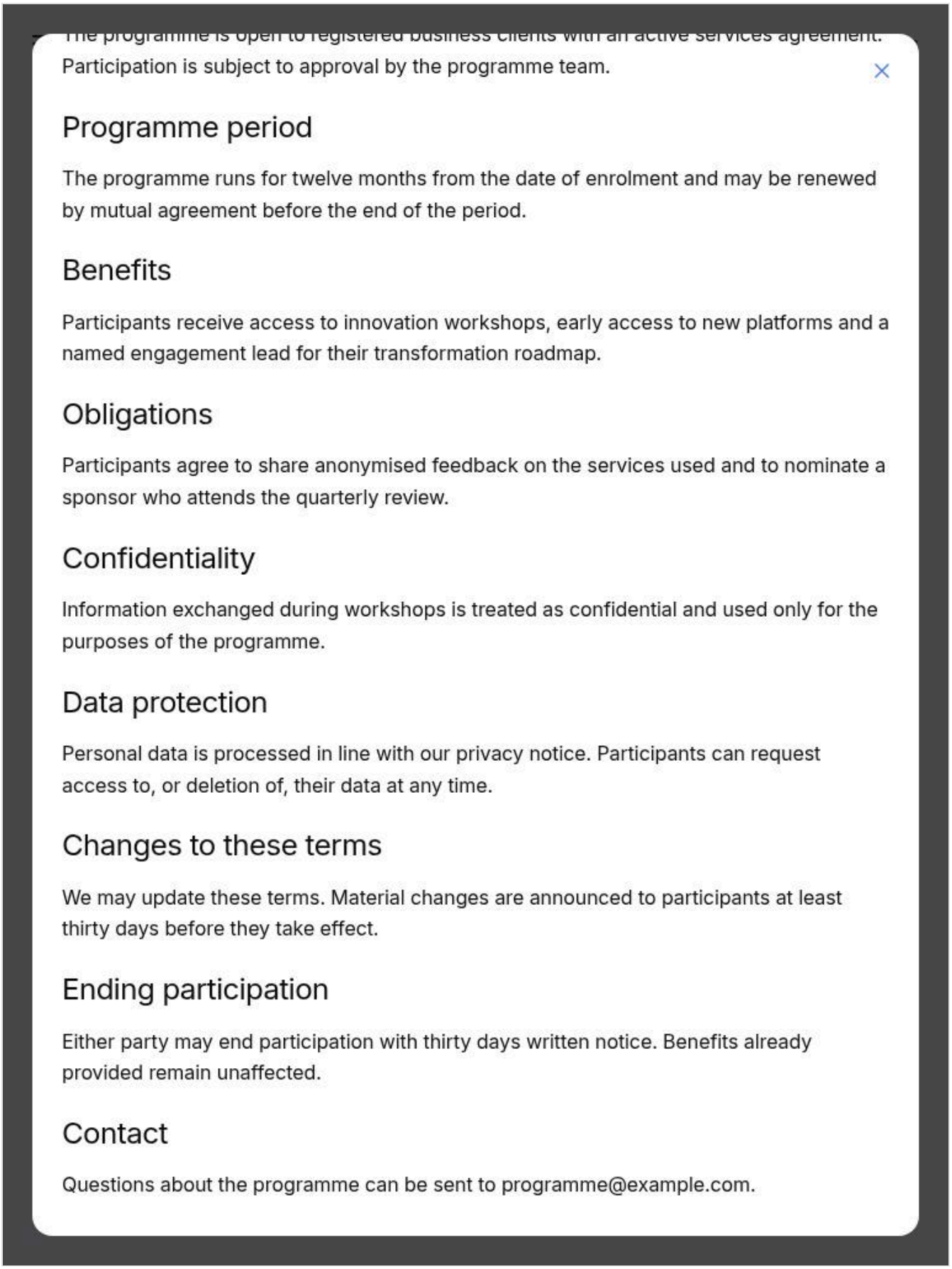


Figure: Tablet rendering — At 768 px, after selecting Read the programme terms and scrolling inside the dialog: a white dialog 1.5rem from each screen edge over the dark backdrop, showing the middle of the programme terms (Heading 3 sections with their paragraphs), with the close (X) button still at the top right.

(Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/modal-tablet-view.jpg) (handover screenshot; rendering at the time it was taken).

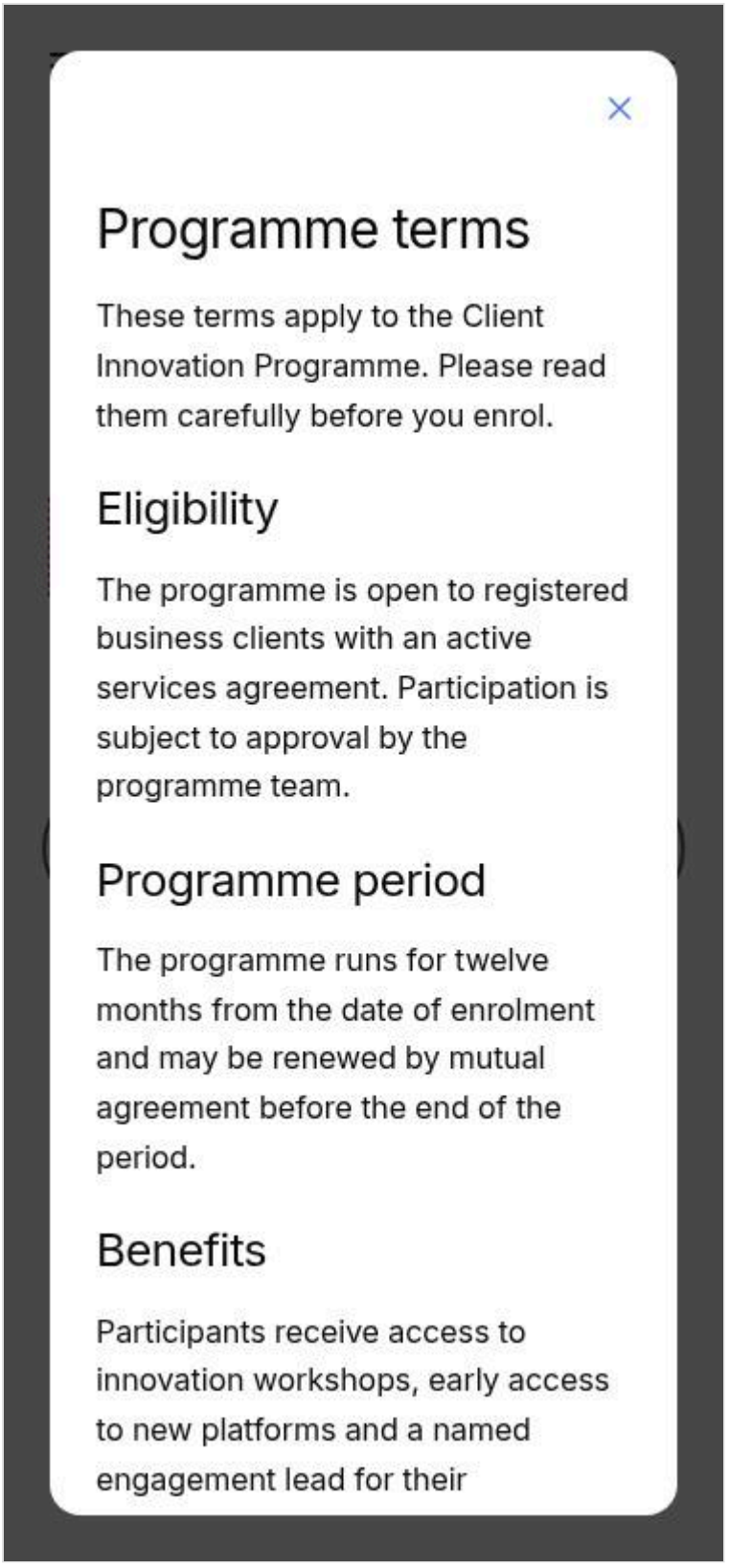


Figure: Mobile rendering — At 375 px, after selecting Read the programme terms: the dialog 1.5rem from each screen edge and as tall as the screen minus 3rem, showing the close (X) button at the top right, the H2 Programme terms, the introduction and the first Heading 3 sections; the rest scrolls inside. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/modal-mobile-view.jpg) (handover screenshot; rendering at the time it was taken).

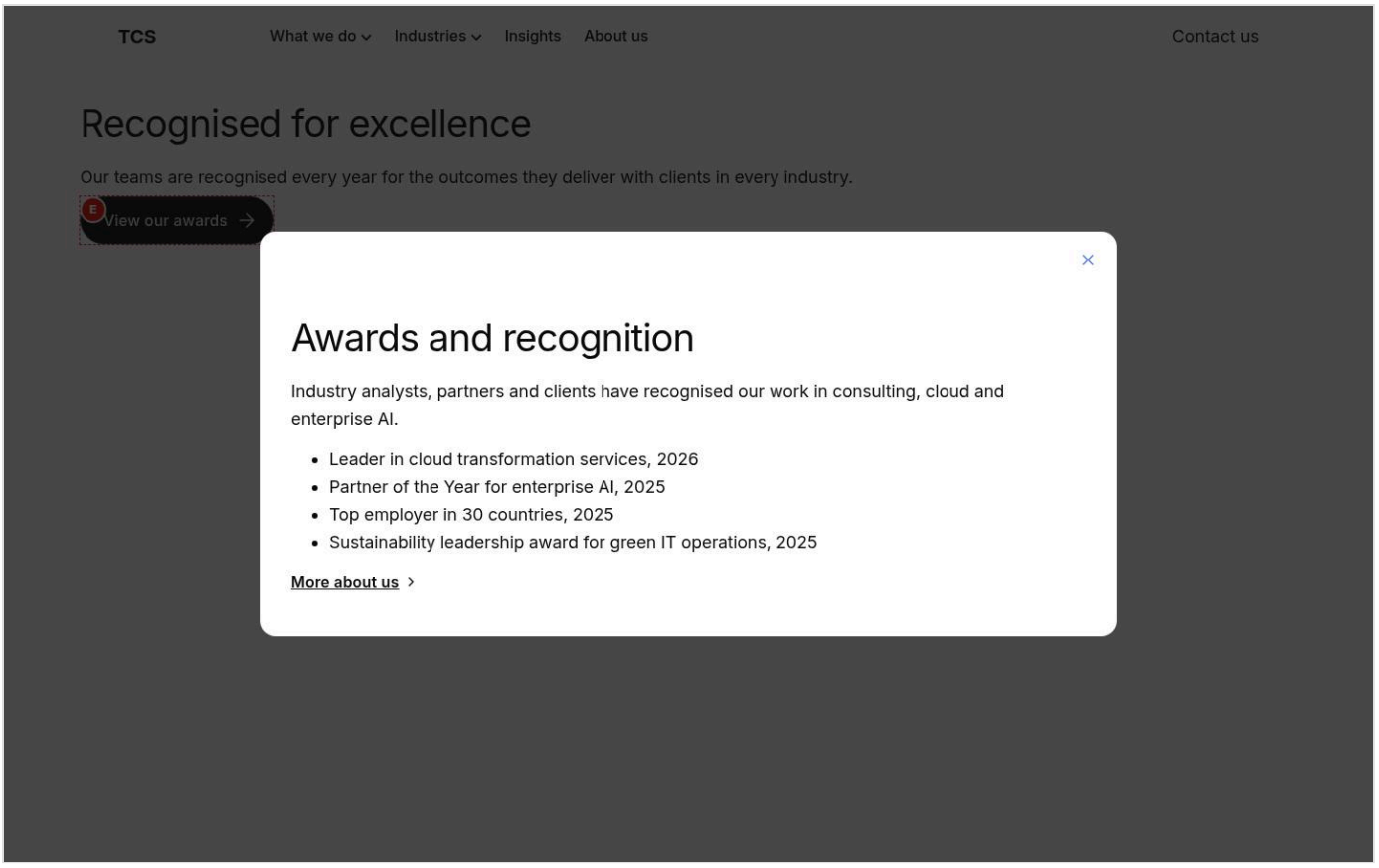


Figure: Modal block variation — At 1440 px: the section with the H2 Recognised for excellence, the paragraph and the primary button View our awards, dimmed under the dark backdrop, and the awards dialog (56.25rem wide, the H2 Awards and recognition, four bulleted awards and More about us) open over it rather than next to it. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/modal-block-button.jpg) (handover screenshot; rendering at the time it was taken).

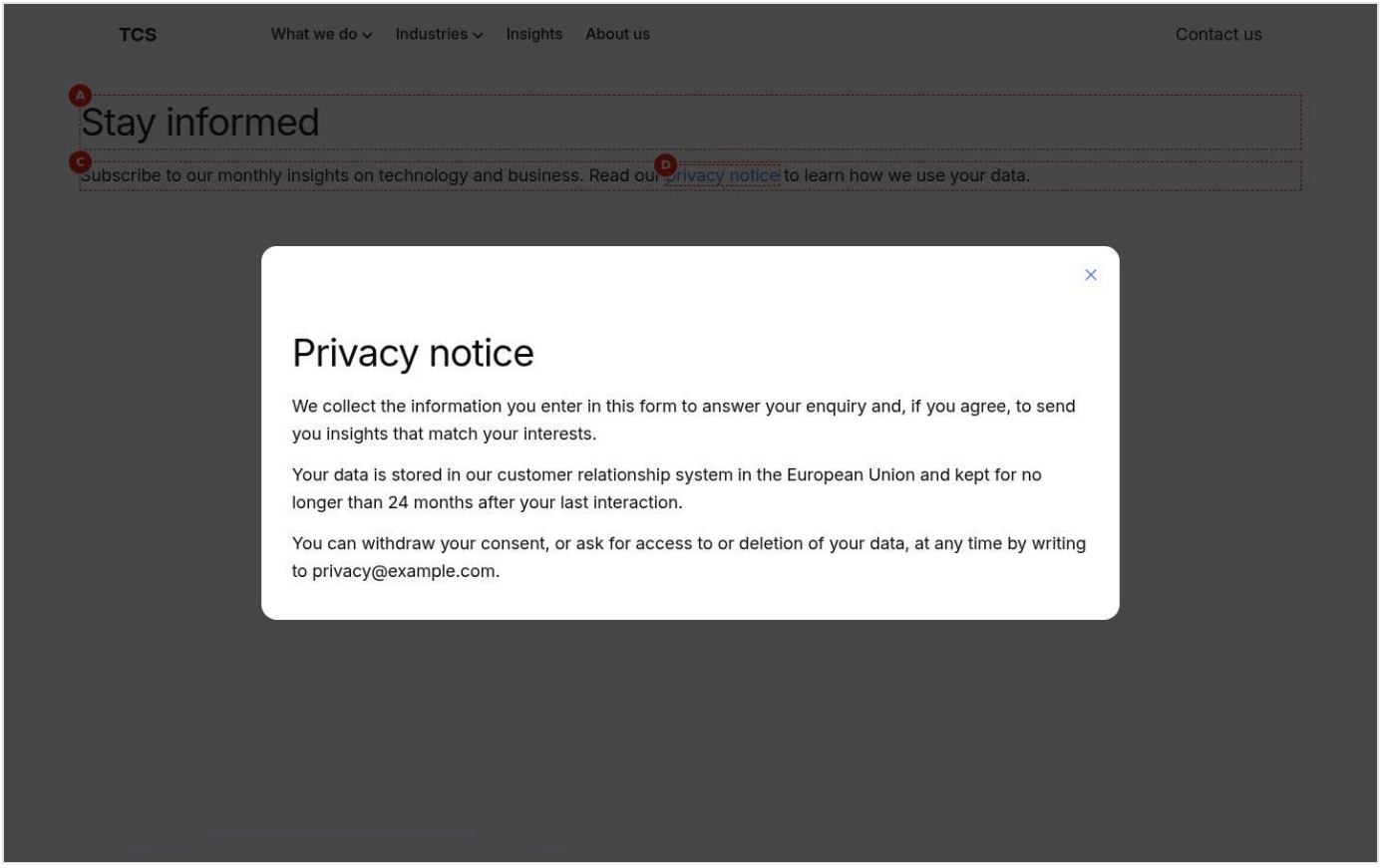


Figure: Auto-linked modal variation — At 1440 px: the H2 Stay informed and the paragraph with the inline link privacy notice, dimmed under the backdrop, and over them the dialog opened by that link with the close button, the H2 Privacy notice and three paragraphs. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/modal-auto-link.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Universal Editor canvas — Universal Editor edit mode at 1440 px: the section with the H2 Recognised for excellence, the paragraph and the primary button View our awards; letter A marks the Modal block and B its opener button. No dialog is open. (Annotated preview in editor mode with sample content; letters as in the annotations list, not a Universal Editor screenshot; file screenshots/modal-authoring-view.jpg) (handover screenshot; rendering at the time it was taken).

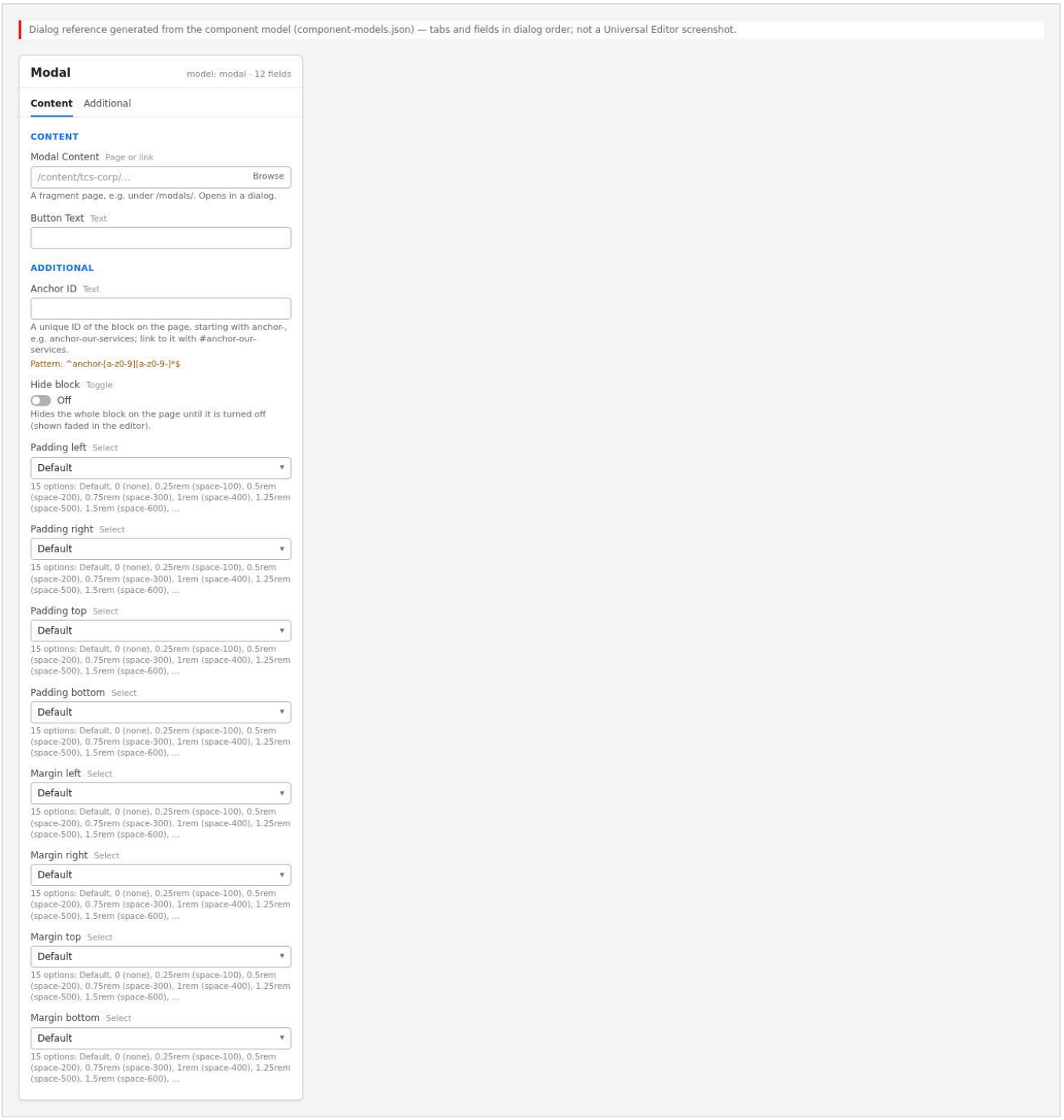


Figure: Properties panel — The Modal properties panel with the Content tab (Modal Content picker and Button Text) and the Additional tab. (Dialog reference generated from the component model, not a Universal Editor screenshot; file screenshots/modal-properties-panel.jpg) (handover screenshot; rendering at the time it was taken).

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages